F.IT309 – Мобайл технологи

Бэлтгэсэн багш: Н.Ганхуяг

Flutter гэж юу вэ?

IOS болон Android-д зориулсан Native аппликешн бүтээх UI фрэймворк юм.

Олон төрлийн төхөөрөмжүүдэд зориулан нэг аппликешнийг нэг кодоор үүсгэх боломжтой.

Fantastic developer tools | easy-to-use language

Flutter 3.0

- Open source framework by Google
- Multi platform
- Fast performance
- Fast development

Flutter 1.0 (Only mobile development)

Flutter 2.0 (Null safety, Web, Desktop, Embedded)

Flutter 3.0 (Material 3, Stable multi platform)

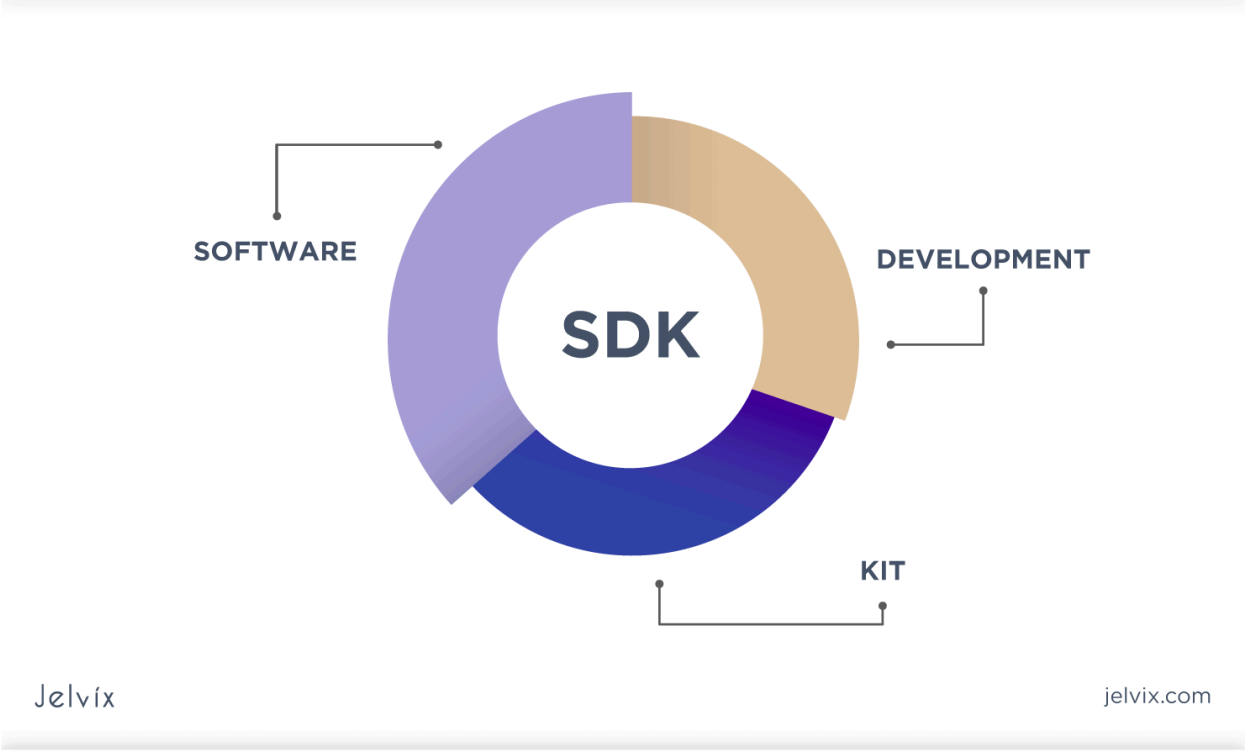
Яагаад Flutter?

- Сурахад хялбар (Dart programming language)
- Хамгийн алдартай мобайл апп хөгжүүлэх платформ
- Хамгийн хурдацтай хөгжиж буй фрэймворк
- Play store 80% app - flutter

Flutter

Highly Customized
User Experiences
Fast
Development
Morden Reactive
framework
Single
Codebase
SDKs & Native
Features Access
Cross Platform Mobile UI- Flutter

SDK?



android SDK

Google Cloud SDK

Google Cloud

Android SDK & IOS SDK

Flutter SDK cross-platform

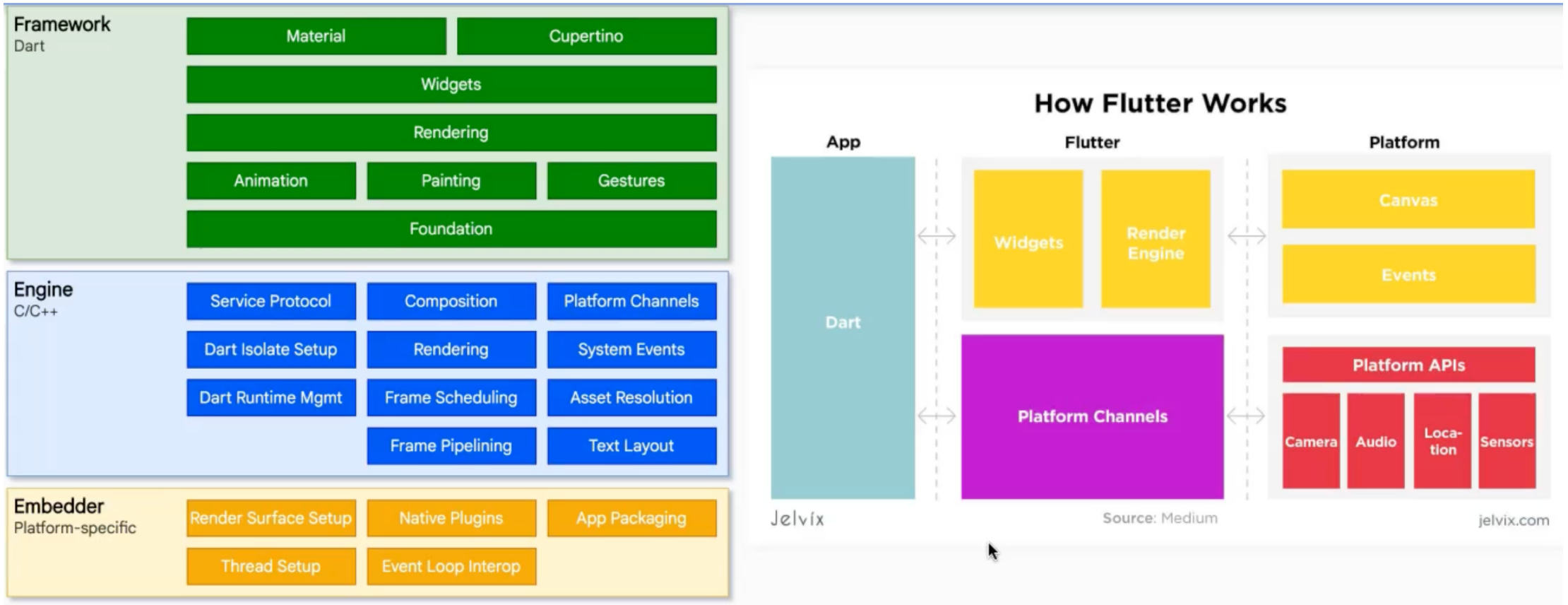
- Xamarin – 2011
- Ionic – 2013
- React native – 2015
- Flutter - 2017

	★★★★	★★	★★★	★★★★★
Performance	Native-like Experience	Native-like Experience	Native-like Experience	Native-like Experience & High Speed
Programming language	Javascript, Swift, Obj-C, Java	HTML5, CSS, JS	C#, .Net	Dart (C/C++, Java)
Compatible Platforms:	Android 4.1+ and iOS 8+	Android 4.4+, iOS 8+, and Windows 10	Android 4.0.3+, iOS 8+, and Windows 10	iOS 8+ and Android Jelly Bean+
User Interface:	Uses native UI controls	HTML, CSS	Uses native UI controls	Uses proprietary widgets & Top-class UI
Pricing	Open-source framework	Open-source framework and Paid as well.	Open-source framework and Paid as well.	Open-source framework

Flutter SDK

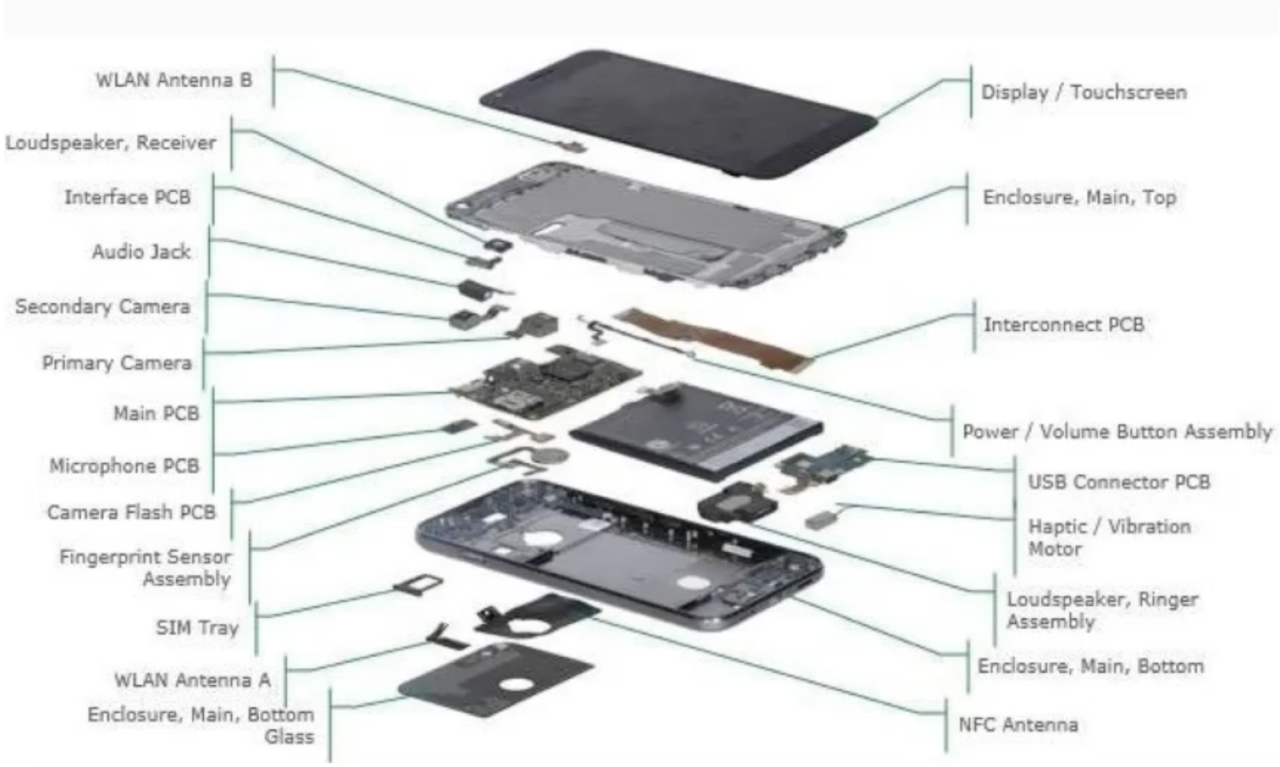
- Dart virtual machine
- Just-in-time execution engine
- Google skia graphics library
- Render хийх давуу тал

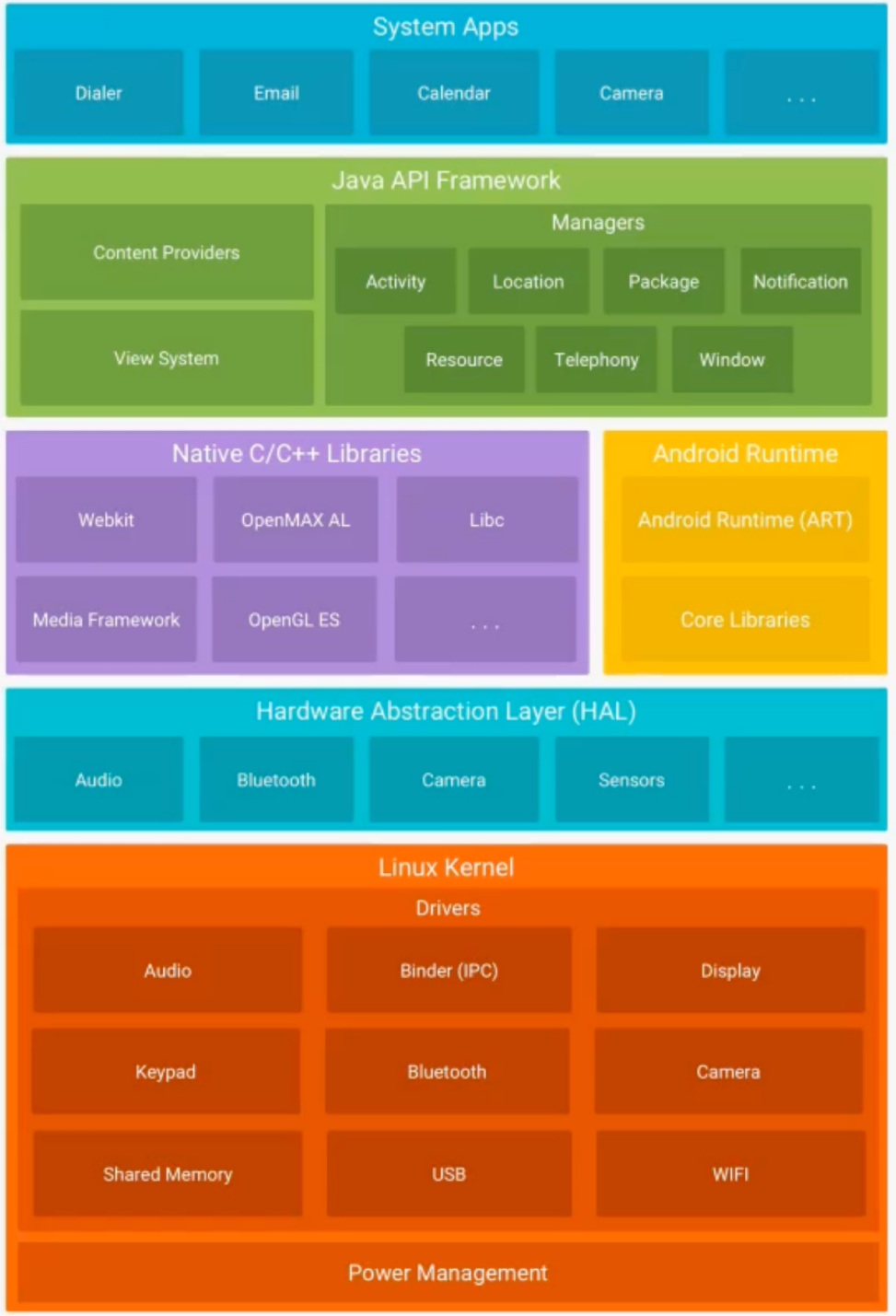
Flutter SDK


Framework
Dart
Material
Cupertino
Widgets
Rendering
Animation
Painting
Gestures
Foundation
Engine
C/C++
Service Protocol
Composition
Platform Channels
Dart Isolate Setup
Rendering
System Events
Dart Runtime Mgmt
Frame Scheduling
Asset Resolution
Frame Pipelining
Text Layout
Embedder
Platform-specific
Render Surface Setup
Native Plugins
App Packaging
Thread Setup
Event Loop Interop
How Flutter Works
App
Flutter
Platform
Dart
Widgets
Render Engine
Canvas
Events
Platform Channels
Platform APIs
Camera
Audio
Loca-tion
Sensors
Jelvix
Source: Medium
jelvix.com

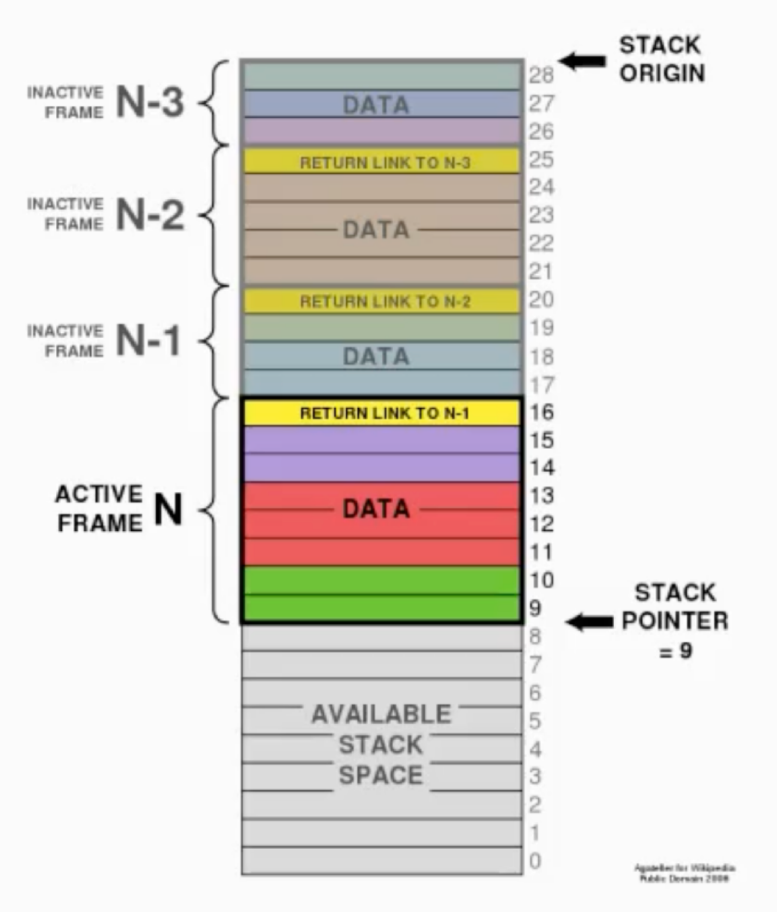
Мобайл төхөөрөмж түүний ялгаатай тал:

Мобайл апп ажиллах зарчим

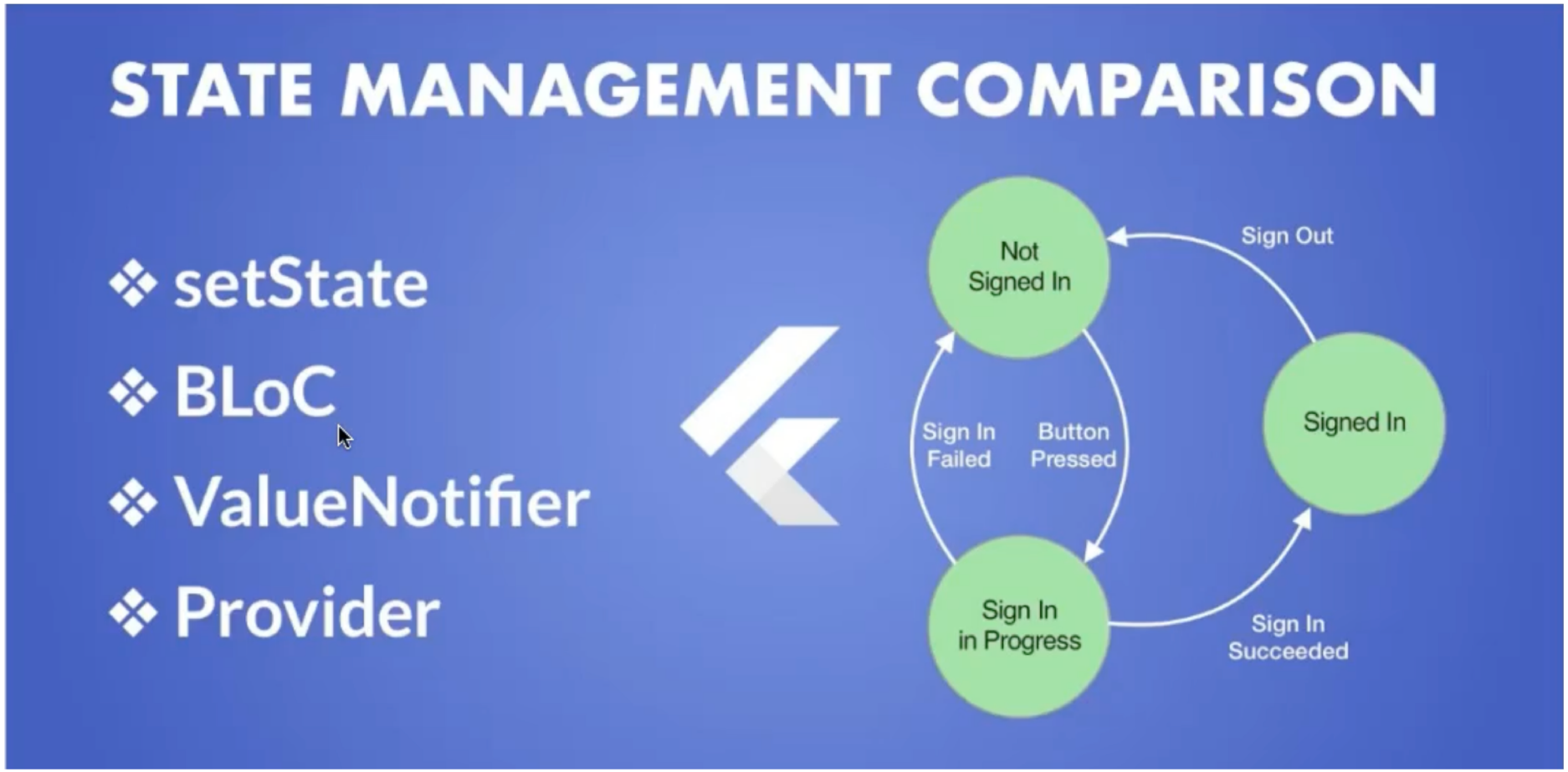




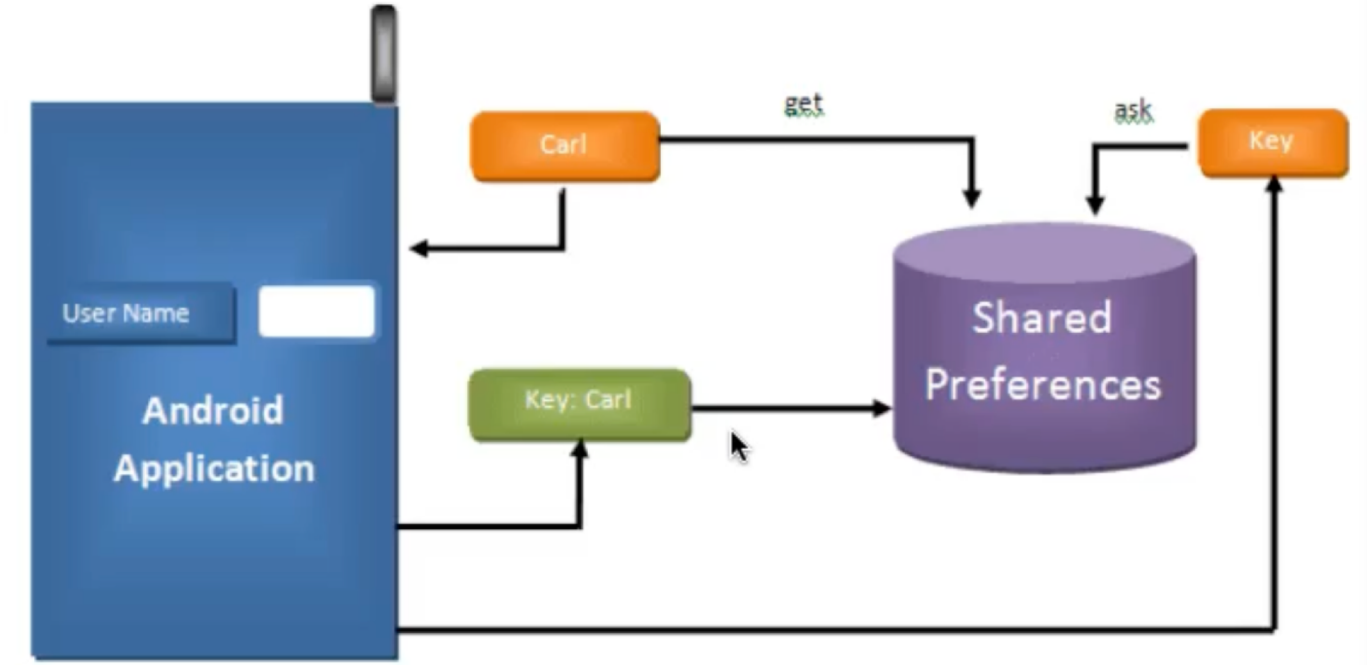
Санах ой ашиглалт

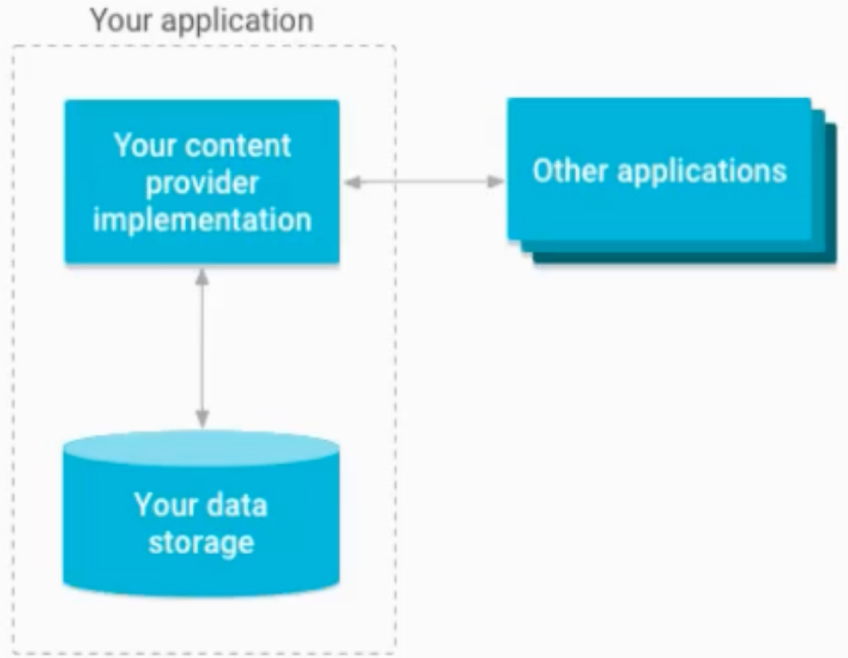


Шуурхай санах ой



Байнгын санах ой

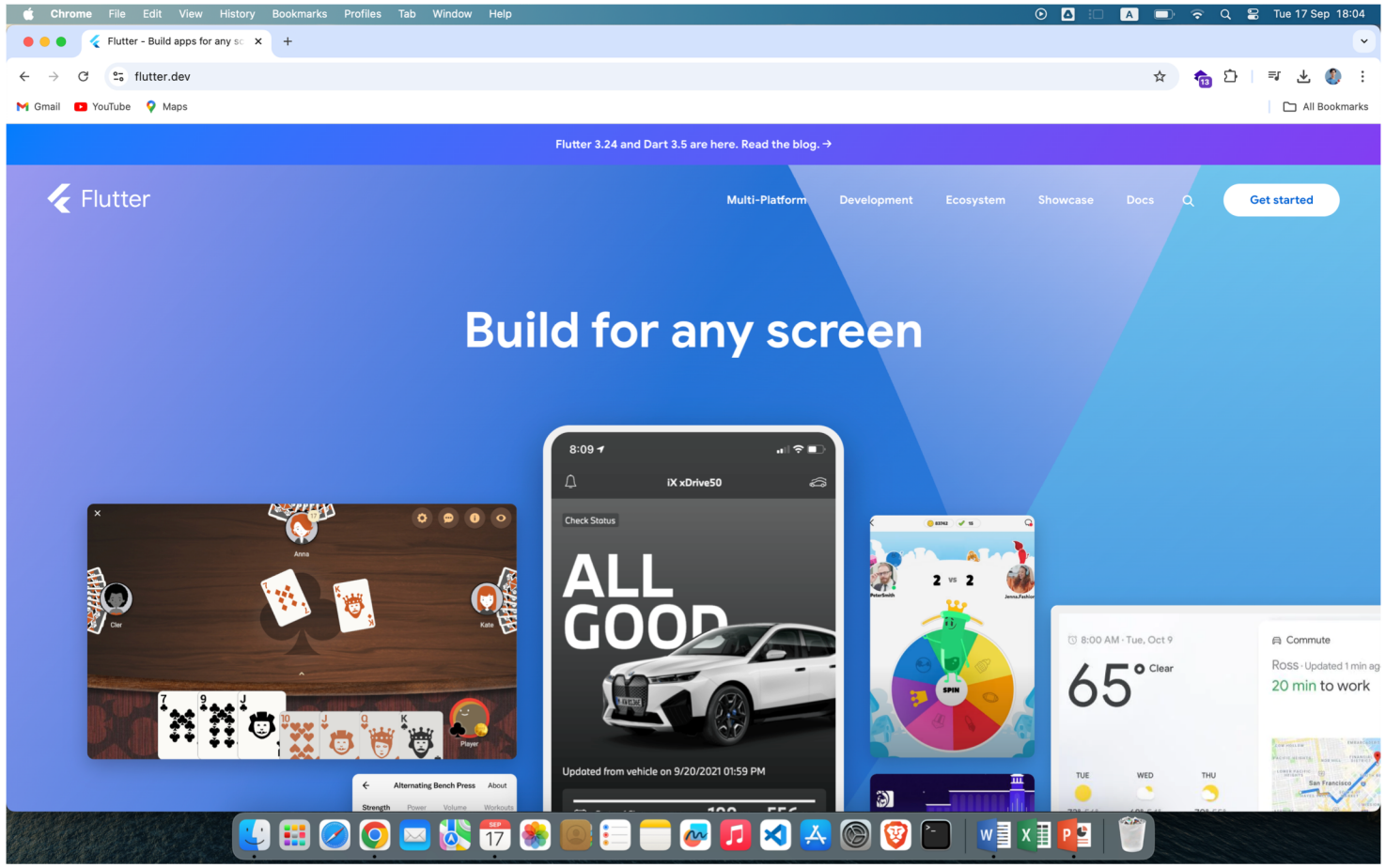




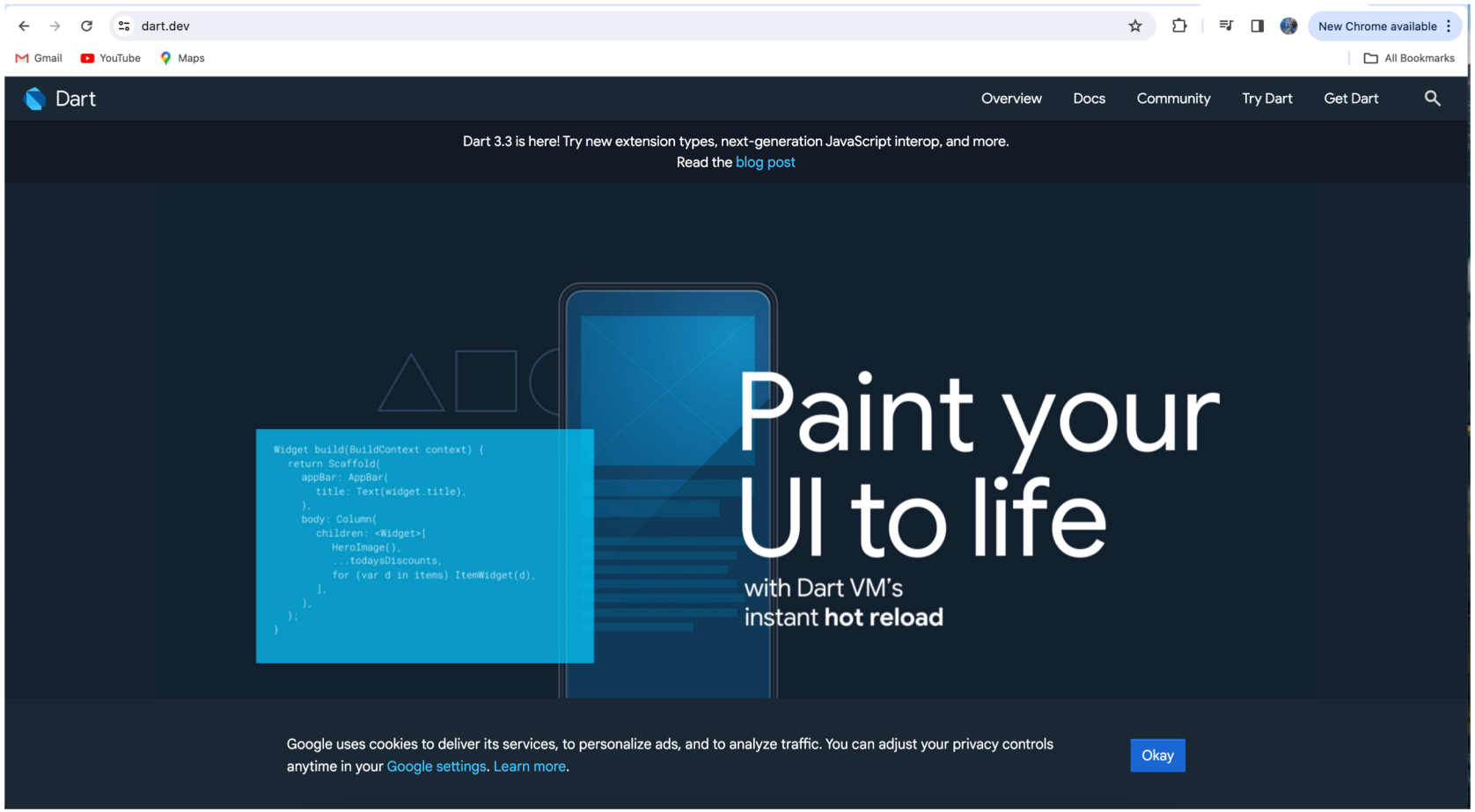
Ямар апп хурдан ажилладаг вэ?

1. Render - UI
2. State management
3. Data, Memory management

flutter.dev


Chrome File Edit View History Bookmarks Profiles Tab Window Help
Tue 17 Sep 18:04
Flutter - Build apps for any sc
flutter.dev
Gmail YouTube Maps
All Bookmarks
Flutter 3.24 and Dart 3.5 are here. Read the blog. →
Flutter
Multi-Platform
Development
Ecosystem
Showcase
Docs
Get started
Build for any screen
8:09
iX xDrive50
Check Status
ALL GOOD
Updated from vehicle on 9/20/2021 01:59 PM
2 vs 2
SPIN
8:00 AM · Tue, Oct 9
65° Clear
Commute
20 min to work
TUE
WED
THU
Alternating Bench Press
About

dart.dev


dart.dev
New Chrome available
Gmail
YouTube
Maps
All Bookmarks
Dart
Overview
Docs
Community
Try Dart
Get Dart
Dart 3.3 is here! Try new extension types, next-generation JavaScript interop, and more.
Read the blog post
Widget build(BuildContext context) {
return Scaffold(
appBar: AppBar(
title: Text(widget.title),
),
body: Column(
children: <Widget>[
HeroImage(),
...todaysDiscounts,
for (var d in items) ItemWidget(d),
],
),
);
}
Paint your UI to life
with Dart VM's instant hot reload
Google uses cookies to deliver its services, to personalize ads, and to analyze traffic. You can adjust your privacy controls anytime in your Google settings. Learn more.
Okay

- АНХААРАЛ ТАВЬСАНД БАЯРДЛАЛАА